

Dr. Aleksandra Olejak

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Research positions

- Oct. 2023 – Sept. 2026  **Max Planck Institute for Astrophysics**, Postdoctoral Fellow.
- Oct. 2019 – Sept. 2023  **Nicolaus Copernicus Astronomical Center, the Polish Academy of Sciences**, PhD student.
- Oct. 2018 – Sept. 2019  **Center of Theoretical Physics, the Polish Academy of Sciences**. A scholarship student under the MAESTRO research grant (PI B. Czerny).

Education

- Oct. 2019–Sept. 2023  **Nicolaus Copernicus Astronomical Center, Warsaw**. Ph.D. degree (Cum laude), thesis defense on Sept. 29th, 2023.
- 2017–2019  **University of Warsaw, Department of Physics**. M.Sc. in Astronomy.
- 2013–2017  **Warsaw University of Technology, Department of Physics**. B.Sc. in Applied Physics.

Selected talks and seminars

- August 2026  Florence, Italy; [Upcoming: **Invited Talk**] *Black-hole spins - measurement techniques and results* during conference *COSPAR 2026*.
- June 2026  Potsdam, Germany; [Seminar] *The origin of binary black hole mergers revealed by LIGO–Virgo–KAGRA: are we beginning to see multiple populations?* seminar at the Astrophysical and Cosmological Relativity division of the Max Planck Institute for Gravitational Physics.
- April 2026  Heidelberg, Germany; [**Invited Talk**] *The Puzzling Formation of Black Hole Systems Revealed by GAIA* during the workshop *Multiple Systems Across Scales*.
- April 2026  Bern, Switzerland; [Review Talk] *Spins of Stellar Black Hole Why do we expect them to be low?* during ISSI workshop *What are the spins of stellar-mass black holes?*.
- October 2025  Munich, Germany; [**Invited Talk**] *Gravitational Waves from Stars Stripped by Super-massive Black Holes* during conference *ORIGINS Black Hole Days*.
- October 2025  Paris, France; [Talk] *The Origin and Formation Channels of GW sources - Current Status* during the meeting *Journées de l'ATPEM 2025*.
- September 2025  Catania, Italy; [**Invited Talk**] *Formation Pathways of GW Sources: Isolated Binary Evolution vs. Dynamical Channels* during conference *Crossroads in Strong Gravity Challenges & Future Directions*.
- September 2025  Oxford, England; [**Invited Talk**] *The Origin and formation channels of LVK gravitational wave sources: current status* during conference *5th Philip Wetton Workshop*.

Selected talks and seminars (continued)

- June 2025  Madrid, Spain; [Talk] *Supermassive black hole with a stripped subgiant companion as a GW source detectable by LISA during conference X-ray Quasi-Periodic Eruptions & Repeating Nuclear Transients*
- June 2025  Princeton, USA; [Seminar] *The origin of gravitational wave sources. Isolated binary progenitors of binary black hole mergers.* Informal seminar for the research group of prof. Eliot Quataert at Princeton University, Department of Astrophysical Sciences.
- April 2025  Milano, Italy; [Seminar] *Supermassive black holes stripping a subgiant star down to its helium core: a new type of multi-messenger source for LISA.* Seminar for the astrophysics department of Milano Bicocca.
- March 2025  Munich, Germany; [Talk] *Supermassive black holes stripping a subgiant star down to its helium core: a new type of multi-messenger source for LISA.* Talk during MIAPbP workshops *Unveiling a Universe of Black Holes: The next Generation of AGN Surveys.*
- December 2024  Paris, France; [Seminar] *The origin of compact object mergers.* Seminar for high energy group of the Institut d'Astrophysique de Paris.
- September 2024  Garching bei Munchen, Germany; [Talk] *Gravitational wave signal from circular, mass transferring star- supermassive black hole systems during conference LISA Astrophysics Working Group Meeting 2024.*
- October 2024  Leiden, Netherlands; [**Invited Talk**] *The origin of compact object mergers. Isolated binary formation scenarios.* during conference *Challenges and future perspectives in gravitational-wave- wave astronomy: O4 and beyond.*
- September 2024  Warsaw, Poland; [Talk] *Gravitational wave signal from circular, mass transferring star- supermassive black hole systems during conference Galactic and extragalactic X-ray transients, theory and observational perspectives.*
- August 2024  Cape Town, South Africa; [Talk] *Unequal-mass, highly-spinning binary black hole mergers in stable mass transfer formation channel during IAU General Assembly 2024.*
- July 2024  Liege, Belgium; [Talk] *Unequal-mass, highly-spinning binary black hole mergers in stable mass transfer formation channel during 41st Liège International Astrophysical Colloquium: The eventful life of massive star multiples.*
- May 2024  Birmingham, UK; [**Invited Panelist**] *Panel Discussion - Binary Formation Channels during Gravitational Wave Physics and Astronomy Workshop GWPAW 2024.*
- March 2024  Warsaw, Poland; [Seminar] *Fingerprints of binary star interactions in the parameters of binary black hole mergers - department seminar at Warsaw Astronomical Observatory.*
- November 2023  Garching bei Munchen, Germany; [Seminar] *The origin of gravitational wave sources. Isolated binary progenitors of binary black hole mergers - department seminar at Max Planck Institute of Astrophysics.*
- July 2023  Milano, Italy; [Panelist] *What can/should astrophysicists and pop-synthers predict? during Gravitational-wave populations: what's next?*
- Jan. 2023  Garching bei Munchen, Germany; [Seminar] *What can we learn about stars from gravitational waves? - seminar at Max Planck Institute for Astrophysics during Stellar Department Meeting.*
- Jan. 2023  Quy Nhon, Vietnam; [**Invited Talk**] *What can we learn about stars from gravitational waves? during Rencontres du Vietnam. TMEX-2023 Conference.*
- April 2022  San Juan, Puerto Rico; [Talk] *The role of supernova convection for the depth of lower mass gap and the isolated binary formation of LVK sources during 2022 INTERMEDIATE-MASS BLACK HOLES New Science from Stellar Evolution to Cosmology.*
- Nov. 2021  Torun, Poland; [Seminar] *Impact of common envelope development criteria on the formation of BH-BH mergers - departmental seminar at Nicolaus Copernicus Astronomical Center in Torun.*

Selected talks and seminars (continued)

- Nov. 2021 📖 Online; [Seminar] *The origin of BH-BH mergers. The isolated binary evolution scenario.* Seminar for high energy group of the Institut d'Astrophysique de Paris.
- Sept. 2021 📖 Online; [Talk] *Impact of common envelope development criteria on the formation of LIGO/Virgo sources* during XL Congress of the Polish Astronomical Society 2021.
- Aug. 2021 📖 Lisbon, Portugal; [Talk] *Impact of common envelope development criteria on the formation of LIGO/Virgo sources* during Global meeting of the GWVerse COST action Lisbon.
- July 2021 📖 Online; [Talk] *Binary neutron star formation and the origin of GW170817* during Sixteenth Marcel Grossman Meeting.
- Sept. 2019 📖 Warsaw, Poland; [Talk] *Synthetic catalog of black holes in the Milky Way* during Astrophysics with GW detections.

Summer schools

- Aug. 2022 📖 Sanata Barbara, USA. Participation in *The 2022 MESA Summer School*.
- Aug. 2021 📖 Copenhagen, Denmark. Participation in *NBIA Summer School on Gravitational Wave Astrophysics*.

Awards and funding

- Jun 2025 📖 Winner of **Procope Mobility** scholarship for 1-month research visit sponsored by the French Embassy in Germany¹.
- Feb 2025 📖 Winner of **SALTO** exchange program between MPG, Germany, and CNRS, France.
- July 2024 📖 **Lindau Nobel Laureate Meeting** participant (1 of ~ 650 selected globally, 1 of 8 in Poland)
- July 2022 📖 Winner of scholarships from **the Minister of Education and Science (Poland)** for outstanding young scientists in 2022.
- May 2022 📖 Winner of **START 2022** stipend of the Foundation for Polish Science dedicated to 100 best young scientists (under 30 year old) that represent all fields of science with additional honor award of **the Prof. Adam Sobiczewski Prize** for researchers who conduct scientific research in the field of mathematics, theoretical physics, or astronomy.
- Dec. 2021 📖 Winner of **Award for young researcher 2021** of Nicolaus Copernicus Astronomical Center.

Skills

- Programming 📖 C, C++, Python, Bash, LaTeX,
- Other tools 📖 Overleaf, Population synthesis codes, MESA
- Language 📖 Polish (native), English (excellent), French (beginner), German (beginner)

Additional activities

Duties and roles

- 📖 Team member (by invite) of the International Space Science Institute project titled: *Can we be spin doctors and determine the spins of black holes?* led by prof A. Zdziarski.
- 📖 Core member of LISA Consortium (since May 2025).

¹declined for personal reasons

Additional activities (continued)

- Postdoc representative at Max Planck Institute for Astrophysics (since June 2024).
- Part of the Stellar Department's recruitment committee in selecting PhD candidates for IMPRS) program
- Referee for journals: Nature, Nature Communications, Nature Astronomy, MNRAS, ApJ, ApJL, A&A
- PhD students representative in the Scientific Council (2021/22) of the Nicolaus Copernicus Astronomical Center, Warsaw.

Organization

- Part of Scientific Organizing Committee of the conference: *Many Faces of Black Holes* (upcoming) <https://sites.google.com/view/bh-nepal-2026/home>
- Member of the Organizing Committee of the Joint Ringberg Conference for Post-docs of the Max Planck Institute for Astrophysics and the Max Planck Institute for Extraterrestrial Physics.
- Part of Local Organizing Committee of LISA Astrophysics Working Group Meeting at Max Planck Institute for Astrophysics: <https://events.gwdg.de/event/875/overview>
- Co-organized Stellar Department Seminars (SESTAS) at Max Planck Institute for Astrophysics (2023/24)
- Organizer and main coordinator of *5th Meeting of young astronomers 2022* at Nicolaus Copernicus Astronomical Center (online). Event website: <https://events.camk.edu.pl/event/41/>.

Teaching

- Supervisor of bachelor's thesis at Technische Universität München (TUM) titled *Intermediate-Mass Black Hole - Star Systems as Potential Multimessenger Sources for the LISA Detector* by Ismaël Brunet Pac
- Teaching Assistant during MESA workshops being a part of *IAU General Assembly 2024, Cape Town* (IAU Hackathons).
- Physics and Computer Science teacher for middle and high school students during two editions of the school under the sails program, Niebieska Szkoła, specifically the 14th edition (September-November 2016) and the 24th edition (January-March 2020.)

Science popularization

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| August 2025 | ■ AAS Nova highlight for AAS's peer-reviewed journals summarizing my publication by <i>Supermassive Black Holes and Stripped Subgiants: Significant Signals for Future Gravitational Wave Detectors</i> . |
| August 2025 | ■ My Monthly Highlights of MPI for Astrophysics titled <i>Gravitational Waves from Stars Stripped by Supermassive Black Holes?</i> |
| April 2025 | ■ Phys.org and Universe Today . Popular science articles summarizing my publication by <i>Phys.org</i> and <i>Universe Today</i> titled <i>Supermassive black holes could strip stars down to their helium cores</i> |
| April 2025 | ■ Science News Today Popular science articles summarizing my publication <i>Science News Today</i> titled <i>Quiet Deaths: Helium Stars Spiraling into Supermassive Black Holes</i> |
| Oct. 2022 | ■ Magazine ACADEMIA . Article for ACADEMIA, the Magazine of Polish Academy of Sciences: <i>Space-time ripples</i> |
| July 2022 | ■ Magazine Interia . Interview for Tygodnik Interia: <i>Our Galaxy is expected to host around 150 million black holes</i> |
| June 2022 | ■ Polish Radio . Interview for Polish Radio: <i>Secrets of the black holes formation</i> . |

Additional activities (continued)

- Oct. 2020  **Nature Research Highlights.** Interview for Nature Research Highlights: *The odd couple: how a pair of mismatched black holes formed.*
- Oct. 2020  **Science in Poland.** Interview for Science in Poland website: *Scientists explain how two black holes of hugely different masses collided.*
- Sept., Nov. 2019  **Warsaw, Poland.** Lectures for general public organized by Nicolaus Copernicus Astronomical Center *Black holes in the Milky Way..*

Publications

I am a co-author of **30** peer-reviewed scientific articles and the leading author of **8** of them.

ADS Public Library: <https://ui.adsabs.harvard.edu/public-libraries/UoLMup96RQe2hPXDjU3Mcw>.

ORCID: <https://orcid.org/0000-0002-6105-6492/>.

Peer-reviewed:

First and second author

1. **Olejak A.**, Klencki J., Vigna-Gomez A., de Mink S. E., van Son L., Stegmann J., Ryu T., et al., 2025, accepted for publication in ApJ, *Widening of Binaries via Non-conservative Mass Transfer as a Formation Channel for Gaia Black Hole System*, doi:10.48550/arXiv.2511.10728
2. **Olejak A.**, Stegmann J., de Mink S. E., Valli R., Sari R., Justham S., 2025, ApJL 987 L11. *Supermassive black holes stripping a subgiant star down to its helium core: a new type of multi-messenger source for LISA*, doi:10.3847/2041-8213/ade432
3. **Olejak A.**, Klencki J., Xu X.-T., Wang C., Belczynski K., Lasota J.-P., 2024, A&A, 689, A305. *Unequal-mass highly spinning binary black hole mergers in the stable mass transfer formation channel*, doi:10.1051/0004-6361/202450480.
4. **Olejak A.**, Fryer C. L., Belczynski K., Baibhav V., 2022, MNRAS, 516, 2252. *The role of supernova convection for the lower mass gap in the isolated binary formation of gravitational wave sources*, doi:10.1093/mnras/stac2359.
5. **Olejak A.**, Belczynski K., 2021, ApJL, 921, L2. *The Implications of High Black Hole Spins for the Origin of Binary Black Hole Mergers*, doi:10.3847/2041-8213/ac2f48.
6. **Olejak A.**, Belczynski K., Ivanova N., 2021, A&A, 651, A100. *Impact of common envelope development criteria on the formation of LIGO/Virgo sources*, doi:10.1051/0004-6361/202140520.
7. **Olejak A.**, Fishbach M., Belczynski K., Holz D. E., Lasota J.-P., Miller M. C., Bulik T., 2020, ApJL, 901, L39. *The Origin of Inequality: Isolated Formation of a 30+10 Msun Binary Black Hole Merger*, doi:10.3847/2041-8213/abb5b5.
8. **Olejak A.**, Belczynski K., Bulik T., Sobolewska M., 2020, A&A, 638, A94. *Synthetic catalog of black holes in the Milky Way*, doi:10.1051/0004-6361/201936557.
9. Stegmann J., **Olejak A.**, de Mink S. E., 2025, ApJL, 992, L26. *Resolving Black Hole Family Issues Among the Massive Ancestors of Very High-Spin Gravitational-Wave Events Like GW231123*, doi:10.3847/2041-8213/ae0e5f.
10. Sen K., **Olejak A.**, Banerjee S., 2025, A&A, 696, A54. *X-ray emission from helium star–black hole binaries as probes of tidally induced spin-up of second-born black holes*, doi:10.1051/0004-6361/202553829
11. Banerjee S., **Olejak A.**, Belczynski K., 2023, ApJ, 953, 80. *Symmetry Breaking in Merging Binary Black Holes from Young Massive Clusters and Isolated Binaries*, doi:10.3847/1538-4357/acdd59.

12. Romagnolo A., **Olejak A.**, Hypki A., Wiktorowicz G., Belczynski K., 2022, *A&A*, 667, A55. *Testing the presence of a dormant black hole inside HR 6819*, doi:10.1051/0004-6361/202141992.
13. Fryer C. L., **Olejak A.**, Belczynski K., 2022, *ApJ*, 931, 94. *The Effect of Supernova Convection On Neutron Star and Black Hole Masses*, doi:10.3847/1538-4357/ac6ac9.
14. Czerny B., **Olejak A.**, Rałowski M., Kozłowski S., Martinez Aldama M. L., Zajacek M., Pych W., et al., 2019, *ApJ*, 880, 46. *Time Delay Measurement of Mg II Line in CTS C30.10 with SALT*, doi:10.3847/1538-4357/ab2913.

Other co-authored

1. Stegmann J., Antonini F., **Olejak A.**, Biscoveanu S., Raymond V., Rinaldi S., Flanagan B., 2026, *ApJL*, 1000, L59., *Gravitational-wave Observations Suggest Most Black Hole Mergers Form in Triples*, doi:10.3847/2041-8213/ae52ec
2. Zdziarski A. A., Marcel G., Veledina A., **Olejak A.**, Lancova D., 2026, *New Astronomy Reviews*, 102, 101746 *Spins of black holes in X-ray binaries and the tension with the gravitational wave measurements*, doi:10.1016/j.newar.2025.101746
3. Klencki J., Podsiadlowski P., Langer N., **Olejak A.**, Justham S., Vigna-Gómez A., de Mink S. E., 2026, *A&A*, 706, A296 *A fundamental limit to how close binary systems can get via stable mass transfer shapes the properties of binary black hole mergers* doi:10.1051/0004-6361/202555500
4. Wiktorowicz G., Middleton M., **Olejak A.**, Dashwood-Brown C., Ward M.-M., Ingram A., 2026, *A&A*, 706, A232. *Self-lensing binaries as probes of supernova physic*, doi:10.1051/0004-6361/202557456
5. Vigna-Gomez A., Grishin E., Stegmann J., **Olejak A.**, Popa S. A., Liu B., Rajamuthukumar A. S., et al., 2025, *A&A*, 699, A272. *Prompt Stellar and Binary Black Hole Mergers from Chemically Homogeneous Evolution in Triples*, doi:10.1051/0004-6361/202554680
6. Wang C., Bodensteiner J., Xu X.-T., de Mink S. E., Langer N., Laplace E., Vigna-Gómez A., et al., 2024, *ApJL*, 975, L20. *Stripped Helium Star and Compact Object Binaries in Coeval Populations: Predictions Based on Detailed Binary Evolution Models*, doi:10.3847/2041-8213/ad86b7.
7. Fumagalli G., Romero-Shaw I., Gerosa D., De Renzi V., Kritos K., **Olejak A.**, 2024, *PhRvD*, 110, 063012. *Residual eccentricity as a systematic uncertainty on the formation channels of binary black holes*, doi:10.1103/PhysRevD.110.063012.
8. Prince R., Zajaček M., Panda S., Hryniewicz K., Kumar Jaiswal V., Czerny B., Trzcionkowski P., et al., 2023, *A&A*, 678, A189. *Wavelength-resolved reverberation mapping of intermediate-redshift quasars HE 0413-4031 and HE 0435-4312: Dissecting Mg II, optical Fe II, and UV Fe II emission regions*, doi:10.1051/0004-6361/202346738.
9. Leveque A., Giersz M., Askar A., Arca-Sedda M., **Olejak A.**, 2023, *MNRAS*, 520, 2593. *MOCCA-Survey Database: extra galactic globular clusters - III. The population of black holes in Milky Way and Andromeda-like galaxies*, doi:10.1093/mnras/stad240.
10. Prince R., Zajaček M., Czerny B., Trzcionkowski P., Bronikowski M., Sobrino Figaredo C., Panda S., et al., 2022, *A&A*, 667, A42. *Wavelength-resolved reverberation mapping of quasar CTS C30.10: Dissecting Mg II and Fe II emission regions*, doi:10.1051/0004-6361/202243194.
11. Wang Y., Liao S., Giacobbo N., **Olejak A.**, Gao J., Liu J., 2022, *A&A*, 665, A111. *Astrometric mass measurement of compact companions in binary systems with Gaia*, doi:10.1051/0004-6361/202243684.
12. Belczynski K., Doctor Z., Zevin M., **Olejak A.**, Banerje S., Chattopadhyay D., 2022, *ApJ*, 935, 126. *Black Hole-Black Hole Total Merger Mass and the Origin of LIGO/Virgo Sources*, doi:10.3847/1538-4357/ac8167.

13. Belczynski K., Romagnolo A., **Olejak A.**, Klencki J., Chattopadhyay D., Stevenson S., Coleman Miller M., et al., 2022, ApJ, 925, 69. *The Uncertain Future of Massive Binaries Obscures the Origin of LIGO/Virgo Sources*, doi:10.3847/1538-4357/ac375a.
14. Zajaček M., Czerny B., Martinez-Aldama M. L., Rałowski M., **Olejak A.**, Przyłuski R., Panda S., et al., 2021, ApJ, 912, 10. *Time Delay of Mg II Emission Response for the Luminous Quasar HE 0435-4312: toward Application of the High-accretor Radius-Luminosity Relation in Cosmology*, doi:10.3847/1538-4357/abe9b2.
15. Zajaček M., Czerny B., Martinez-Aldama M. L., Rałowski M., **Olejak A.**, Panda S., Hryniewicz K., et al., 2020, ApJ, 896, 146. *Time-delay Measurement of Mg II Broad-line Response for the Highly Accreting Quasar HE 0413-4031: Implications for the Mg II-based Radius-Luminosity Relation*, doi:10.3847/1538-4357/ab94ae.
16. Belczynski K., Klencki J., Fields C. E., **Olejak A.**, Berti E., Meynet G., Fryer C. L., et al., 2020, A&A, 636, A104. *Evolutionary roads leading to low effective spins, high black hole masses, and O1/O2 rates for LIGO/Virgo binary black holes*, doi:10.1051/0004-6361/201936528.

Other submitted

1. Banerjee S., **Olejak A.**, 2024. *On the effective spin-mass ratio $\chi_{\text{eff}} - q$ relation of binary black hole mergers that evolved in isolation* doi:10.48550/arXiv.2411.15112
2. Belczynski K., Bulik T., **Olejak A.**, Chruslinska M., Singh N., Pol N., Zdunik L., et al., 2018. *Binary neutron star formation and the origin of GW170817*, doi:10.48550/arXiv.1812.10065.